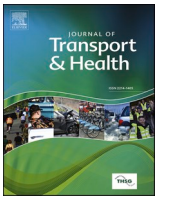
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Public transport for people with hearing impairment: A scoping review

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ABSTRACT

Introduction: Mobility is an important aspect of daily life, yet people with disabilities often encounter barriers when using public transport. In this paper, we reviewed the literature on accessibility in public transport for passengers with hearing impairment and identified several recurring challenges.
Methods: To structure the material, we grouped the findings into four areas: access to information, environmental conditions, vehicle design, and interactions with staff. Within these areas, the studies describe a range of practical measures, such as clearer high-contrast visual signage, real-time information displays, tactile guidance, visual emergency alerts, and more targeted staff training.
Results: Recurring challenges for people with hearing impairments include difficulties accessing important information, barriers in the surrounding environment, limitations in vehicle design, and gaps in staff training. Across studies, these barriers were linked to reduced participation and, in many cases, lower perceived quality of life.
Conclusions: The scoping review points to the need for transport systems that combine universal design principles with digital and assistive technologies and indicates that relatively few studies have examined the specific transport and mobility needs of passengers with hearing impairment. This suggests a need for further research, particularly in relation to how emerging transport systems and services, including automated and digitally supported travel, can be designed to support this group. Addressing the identified issues is likely to improve access, independence, and the mobility of passengers with hearing impairment, and contribute to more inclusive and useable public transport systems.

1. Introduction

Mobility is an important element of everyday life of a working society, particularly for populations at risk of transport-related social exclusion. In this context, accessibility can be seen as the ability to reach valued destinations for various purposes. For individuals, it

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can be access to work, stores, services (such as healthcare) and schools. Approximately 1.3 billion people live with some form of disability, making them the largest minority group on earth, according to the World Health Organisation (WHO, 2023). Disabilities can result from choices about how physical, social, and digital environments are planned and designed. The United Nations (UN) Convention on the Rights of Persons with Disabilities describes this phenomenon as follows: "disability results from the interaction between persons with impairments and attitudinal and environmental barriers that hinder their full and effective participation in society on an equal basis with others." (UN, 2006). Moreover, the Convention on the Rights of Persons with Disabilities is especially interesting to consider in the context of mobility since it has been ratified by 191 states (as of 6/3/2024). Further to that, Article 20 states that all signing states "shall take effective measures to ensure personal mobility with the greatest possible independence for persons with disabilities" (ibid.). However, people with disabilities not only face difficulties with mobility, but also with accessing transportation (Scott et al., 2023). Studies show that people with disabilities travel 30% less frequently than their able-bodied counterparts (K. Park et al., 2023). Beyond mobility itself, this can have significant social impacts, including reduced employment opportunities (Lubin and Deka, 2012), missed medical appointments (Bascom and Christensen, 2017), and feelings of social exclusion (Park and Chowdhury, 2018).

Public transport in many countries and cities is usually a more sustainable and cleaner alternative to cars and plays an important role in policies aimed at improving equitable and inclusive access to mobility. Hence, there is often a strong political and economic motivation to get people to use public means of travel. In Article 9 of the treaty, signing states commit to "enable persons with disabilities to live independently and participate fully in all aspects of life" by taking "appropriate measures to ensure to persons with disabilities access, on an equal basis with others, to the physical environment, to transportation, to information and communications, including information and communications technologies and systems, and to other facilities and services open or provided to the public, both in urban and in rural areas." (UN, 2006). At the same time, transport systems are not fully built to accommodate a range of disabilities, which can reduce service usability and discourage use among vulnerable groups (Guillén et al., 2024). Despite a seemingly global consensus on the importance and ambition of making mobility accessible to all, recent studies show that there are differences in travel patterns and behaviours among people with disabilities (Park et al., 2023).

People with disabilities face very different challenges based on their disabilities. Despite this, transport research often treats people with disabilities as one homogenous group, which can obscure differences in needs, travel experiences, and interaction with transport systems (Figueiredo et al., 2012). In this study, we are especially interested in extant knowledge about how people with hearing impairments experience public transportation, as well as research about how to design services for that group. Hearing impairment is a heterogeneous disability, meaning it varies in severity and affects individuals differently based on factors such as age and genetics. For example, factors that may have an impact are the distance from the speaker, background noise, and available communication methods. These challenges are particularly significant in public transportation, where information is often time-sensitive, multimodal, and dependent on environmental conditions such as noise and crowding. Among people with hearing impairments, an average of 72% make a daily trip, compared to 80% of those without disabilities (Trafikanalys, 2018). To understand the problems faced by people with disabilities, it is important to examine different kinds of barriers. Consequently, previous publications have looked at the barriers people with impairments face (e.g., Rimmer et al., 2004), but also specifically at barriers hindering or preventing people with hearing impairments from using public transport (Bezyak et al., 2017; Fürst and Vogelauer, 2012; Scott et al., 2023). Public transportation systems often fail to provide the support that passengers with hearing impairment need. In many cases, this is due to untrained staff, poor communication systems, and the lack of standardized solutions. Hearing impairment, often considered an "invisible" disability, makes it difficult for transit staff to recognize when a passenger need assistance. As a result, individuals with hearing impairment frequently face barriers such as missing announcements about delays, emergencies, or schedule changes (Bettger and Pearson, 1989; Hunter-Zaworski and Hron, 1999) where information is predominantly delivered through sound. Importantly, the problems may be increased by inconsistent or poorly placed signage and the absence of tactile pathways or other accessible infrastructure, which further hinder navigation and access to information at transit hubs (Starzyńska et al., 2015; Sze and Christensen, 2017).

Limited accessibility in public transport can restrict participation in everyday activities and is associated with poorer quality of life among people with disabilities, including those with hearing impairment (Sze and Christensen, 2017). Addressing these barriers requires attention to how information is communicated and accessed within transport systems, as these factors shape opportunities for participation, independence, and continued use of public transport services. From a wider perspective, access to public transport is thus a question of social sustainability and inclusion. In the transport justice field, correlations have consistently been found between poor access to transportation and social exclusion (Lucas, 2012). In this research area, sustainability includes how opportunities of mobility are distributed among groups of society (Pereira et al., 2017). From that viewpoint, any barriers to participation in public transport can be understood as issues of distributive justice.

In this paper, we provide an overview of the barriers encountered by passengers with hearing impairment in public transport and outline a set of approaches that have been proposed to address them. Based on the reviewed literature, we organise these barriers into four areas: access to information, environmental conditions, vehicle design, and interactions with staff. Across these areas, a range of measures has been described, including clearer visual signage with large fonts and high contrast, real-time information displays, tactile guidance, visual emergency alerts, and more targeted staff training. Taken together, these findings point to the need for transport systems that combine principles of universal design with both established and emerging assistive solutions. They also indicate that accessibility can be considered within a broader shift toward service design and innovation in public transport, including efforts to improve usability, inclusivity, and perceived service quality for vulnerable users. This includes, for example, developments in automated transport systems and digitally supported travel, where visual and multimodal communication may play a more central role.

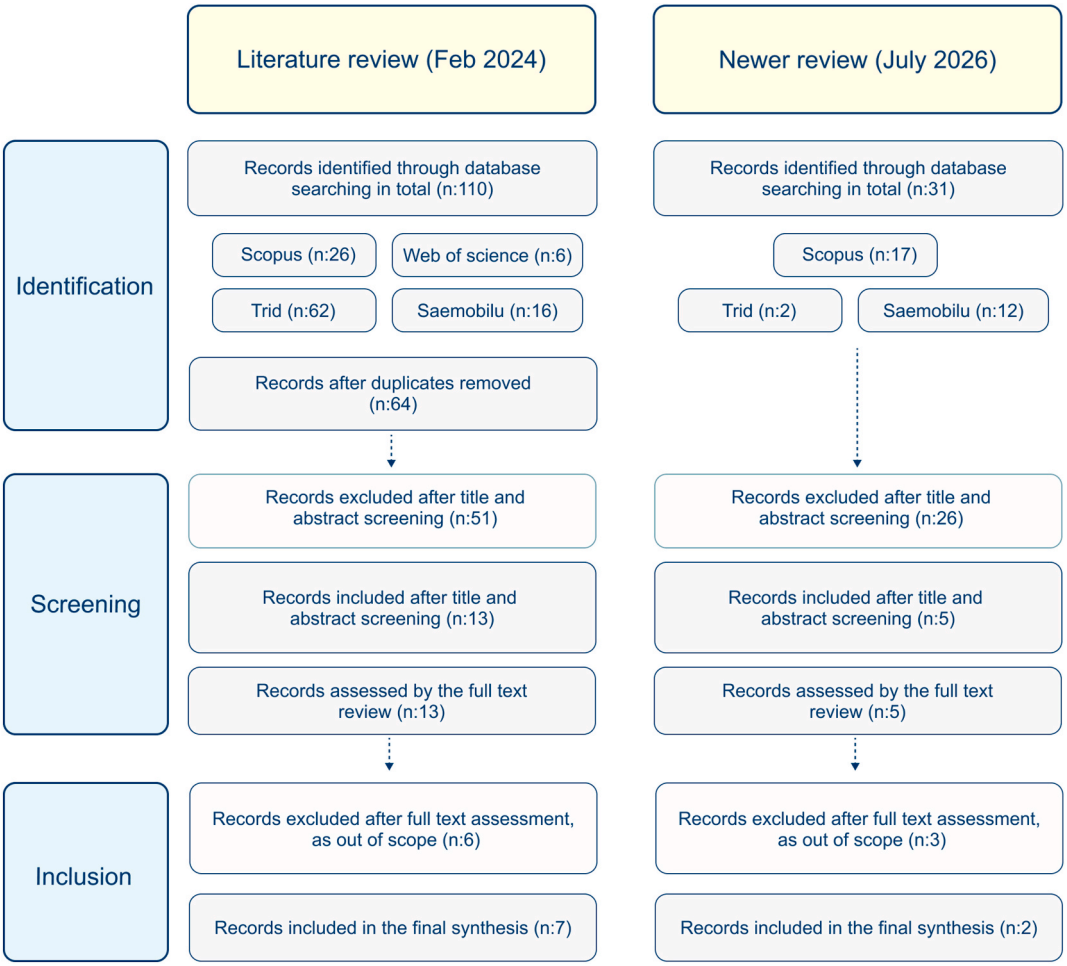


Fig. 1. PRISMA diagram.

Addressing the barriers identified in this review is likely to improve access to transport, support independent travel, and increase participation in everyday activities for passengers with hearing impairment, thereby contributing to more inclusive and useable public transport systems.

2. Method

The literature review was conducted in February 2024 following the PRISMA systematic approach, with a subsequent search conducted July 2026, specifically looking for additional publications between 2024 and 2026. First, we formulated the research question: What is the experience of passengers with hearing impairment when using public transportation? The literature search was carried out using academic databases, including Scopus ((ALL (“public transport*") OR ALL (“mass transit”) OR ALL (“public transit”) AND ABS (“hear × impair*") OR TITLE (“hear × impair*") AND NOT TITLE-ABS-KEY (healthcare) AND NOT TITLE-ABS-KEY (“health care”)) AND (LIMIT-TO (PUBSTAGE, “final")) AND (EXCLUDE (SRCTYPE, “p") OR EXCLUDE (SRCTYPE, “k") OR EXCLUDE (SRCTYPE, “b")), TRID (“hearing impair*"), Web of Science (((((((((ALL=(“public transport*")) OR ALL=(“mass transit")) OR ALL=(“public transit")) AND TS=(“hear × impair*")))) NOT TS=(“health care")))) NOT TS=(Covid)), and Saemobilus (In Metadata only (public) And In Metadata only (hearing impair*)). For the follow-up search, we narrowed down the results to only recent publications and English language.

In the first search, a total of 110 papers were initially retrieved. All papers were then imported into the Mendeley reference management system. After removing the duplicates, the next step was developing inclusion and exclusion criteria. Inclusion criteria were as follows: studies focusing on hearing impairment, auditory disabilities, public transportation, or bus drivers, and articles that were published in English and accessible online. All studies not meeting these criteria, or that were inaccessible, were excluded. In the follow-up search, 31 papers were found (17 Scopus, 2 Trid, 12 Saemobilus) and reviewed.

Table 1
Summary of included papers.

Authors (Year)	Study context	Paper type	Methodology	Theoretical framework	Country studied	Target population	Transportation modes	Key issues	Interventions
Bettger and Pearson (1989)	Public transport accessibility for hearing impaired users	Policy analysis	Qualitative policy analysis, literature review and technology assessment	No	USA	Hearing impaired. Referenced against mobility- and vision-impaired	Bus, subway, airplane	Communication barriers, missed information and safety risks.	Telecommunication devices for the deaf (TDDs), provision of analog information, electronic readerboards, visual emergency alarms, touch screen monitors and staff sensitivity training.
Iannuzziello (2001)	Techniques for communication and information accessibility for person with disabilities	Report	Mixed methods including review, surveys, telephone interviews, expert panel review, and case examples	No.	USA & Canada	Hearing-, visual- and cognitive-impaired & seniors	Bus, heavy rail, light rail, commuter rail, subway, paratransit, ferry	Communication barriers, accessibility gaps, inadequate compliance	Technology based interventions including, visual, auditory, tactile and digital/smart technologies, communication systems, training interventions, design interventions and operational strategies
Fürst and Vogelauer (2012)	Barriers that sight and hearing impaired passengers face when using public transport services	Research paper	Mixed qualitative two-phase approach: expert interviews & associations, w/transport companies, authorities, online discussion forums, and literature review	No.	Austria	People with visual or acoustic impairments	Multi-modal transit - no specific mode mentioned	Communication barriers, information accessibility, Physical/ environmental barriers, navigation issues, safety risks	Information & communication based solutions, technology-based solutions, infrastructure solutions, vehicle related solutions, environmental/ physical solutions, organisational/social interventions
Starzyńska et al. (2015)	Needs and requirements elicitation of people with disabilities and reduced mobility using public transport	Research paper	Focus group interviews.	Journey based model	Poland	Blind/partially sighted, Deaf/ hearing impaired and people with locomotor dysfunction	Bus, tram	Pre-trip, on-trip and post, trip barriers as well as structural problems and system inconsistencies	Information & Communication solutions, technology solutions, infrastructure solutions, operational solutions and user feedback mechanisms
Sze and Christensen (2017)	Review current accessibility practices & guidelines for transport and facilities	Research paper	Qualitative review of: design guidelines & empirical studies based on perception changes & travel behavior	Accessibility design standards and guidelines	USA, UK and Hong Kong	Main: elderly population, Secondary individuals with mobility-, visual- and hearing impairments	Bus, rail and pedestrian infrastructure	Physical/ environmental barriers, design inconsistency, information barriers, safety concerns, and systemic issues	Infrastructure design, transport facility design, vehicle design, information systems design, safety feature improvements, planning and policy solutions and technological solutions.
Tyler (2017)	Interaction and trade-offs between safety, sustainability & accessibility for design of transport systems.	Research paper	Mixed methods: Conceptual framework development, case study analysis, experimental research	Five Cities model, Capabilities model, difference principle	UK	People with reduced mobility, visual- & mobility impairment, and pedestrians	Mainly mixed traffic and pedestrian spaces and infrastructure	Conflicts between sustainability, safety and accessibility, micro-scale barriers, hidden exclusion and cognitive and perceptual issues	Evidence-based and adaptive/inclusive design, policy and planning solutions and equity-based solutions.
Weyrer et al. (2014)	Development of a geospatial barrier catalog, implementation of routing app	Research paper	Mixed methods: data modelling case study, application system development & scenario testing	No	Austria	People with physical, visual and hearing impairments	Public transport (bus, etc.) and walking	Physical/ environmental barriers, infrastructure barriers, data-related barriers and system level issues.	Geospatial barrier catalog, OSM-based routing model, accessible routing algorithm, accesible user interface and scenario-based use cases.

(*continued on next page*)

Table 1 (*continued*)

Authors (Year)	Study context	Paper type	Methodology	Theoretical framework	Country studied	Target population	Transportation modes	Key issues	Interventions
Lee (2025)	Understand social aspects & impact of walking environment for transportation vulnerable	Research paper	Quantitative study consisting of survey and multi-regression analyses	No.	South Korea	People w/ children, physical, visual, or hearing impairment, elderly, & pregnant	City buses (including village buses) and subways	Physical/ environmental barriers, safety issues, psychological/ cognitive factors and risk of social exclusion	Physical improvements to walking environments, pedestrian safety improvement measures, policy recommendations
Alçura (2025)	transportation challenges for people w/ disabilities; perceptions & expectations of service quality based on Sustainable Development Goals (SDGs).	Research paper	Quantitative study consisting of survey and cluster analysis	SERVQUAL	Turkey	People with visual, & hearing impairments, chronic illness & physical disabilities	Public bus	Crowdedness, noise and ventilation, excessive noise, vehicle ergonomics, accessibility to stops, & frequency of service.	Improvments related to infrastructure, service, vehicle, data & planning, environment and safety, and social interventions.

The next step was screening, which was conducted in two stages. First, titles and abstracts were reviewed to exclude irrelevant articles that did not meet the inclusion criteria and second, the full texts of the papers were reviewed, and the papers that focused on hearing-impaired individuals and public transportation were selected. Articles on health care or autonomous vehicles were excluded. In the first search this process narrowed the selection to seven papers that aligned with the research objectives. In the follow-up search, 5 publications met the inclusion criteria, and subsequently 2 were included based on a full paper review, see Fig. 1. Next, the papers were analysed based on the following criteria: they were studying hearing impairment, aging, accessibility, inclusive public transport, ergonomic guidelines, sustainable public transport, and communication strategies.

In the last step general information such as: authors, years of publication, context, country, and target population as well as a summary of findings, were entered into an Excel file for systematic presentation and further analysis (Table 1).

Fig. 1 illustrates the process of identifying, screening, and selecting studies for inclusion in the review. A total number of 141 records were identified through database searches (Scopus, TRID, Web of Science, and Saemobilus). After Identification and Screening, that number was down to a total of 18 publications. These publications were assessed in full text, resulting in 9 studies included in the final synthesis.

3. Findings

The aim of the study was to synthesize the barriers and potential solutions related to the accessibility of public transportation for individuals with hearing impairments. In this section, we present the results organized into the five domains: information accessibility, environmental factors, vehicle design, interactivity, and staff training. Moreover, in this section, we also integrate the findings from the review of available and emerging technologies supporting accessibility, safety, and sustainability in transport systems.

3.1. Barriers categorized by types

3.1.1. Information accessibility

Information accessibility was discussed in three studies, which consistently reported difficulties in accessing information needed to use public transport independently (Starzyńska et al., 2015; Hunter-Zaworski and Hron, 1999; Bettger and Pearson, 1989). These problems included timetables and printed schedules positioned too high or presented in text that was difficult to read, as well as a continued reliance on spoken announcements for delays, schedule changes, and emergency messages. The studies also found that inconsistent signage and unclear route information, both of which made navigation more difficult and reduced access to information needed for independent travel.

3.1.2. Environment

Environmental factors were also reported in three studies, which described several barriers to navigation and information access in transport settings (Fürst and Vogelauer, 2012; Sze and Christensen, 2017) and the quality and safety of the walking spaces (Lee, D., 2025). The factors included the absence of tactile markers or contrasting pathways, which made orientation more difficult, particularly in larger transit hubs and within vehicles. High levels of background noise were also reported to interfere with access to spoken announcements, further limiting their usefulness for passengers with hearing impairment. In addition, the lack of visual alerts for emergencies or vehicle arrivals meant that important updates and instructions could be missed.

3.1.3. Vehicle

One study reported issues with the vehicle's design (Starzyńska et al., 2015). Specifically, the authors found that a major problem was the lack of visual displays that provide real-time information about current and upcoming stops. The lack of visual display resulted in limited access to information. Moreover, Starzyńska et al. (2015) found that certain seating layouts, such as lateral arrangements, could make it more difficult to move within the vehicle and maintain orientation.

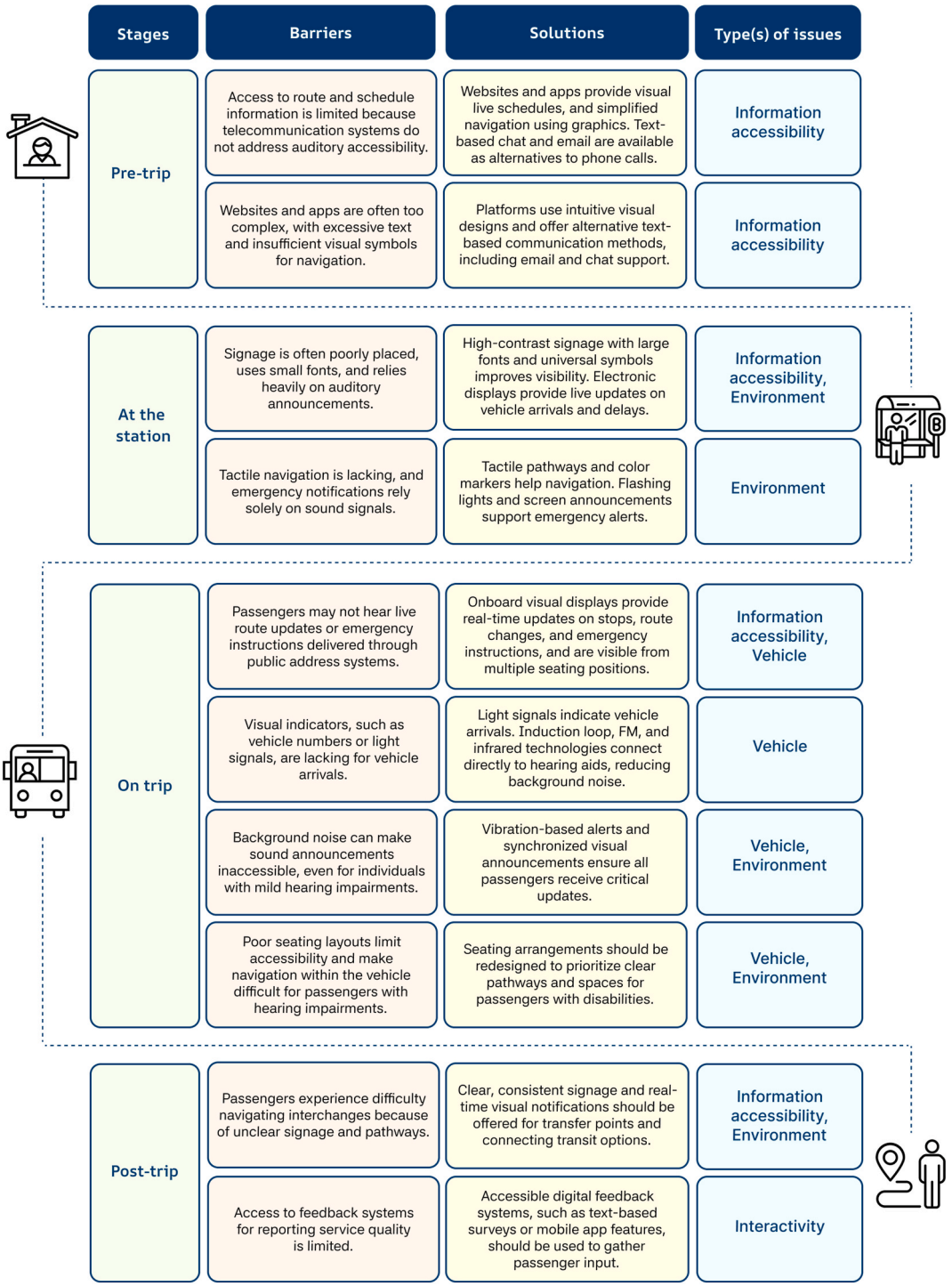


Fig. 2. Overview of barriers and solutions across different stages of the public transport journey.

3.1.4. Interactivity and personnel training

Passengers with hearing impairment may also experience difficulties when seeking information or assistance from transit staff. Two studies pointed to limited staff training as a contributing factor, with personnel often not recognizing when support is needed or lacking strategies for communicating without relying on speech (Bettger and Pearson, 1989; Hunter-Zaworski and Hron, 1999). For example, the use of written communication or basic sign language was rarely reported. These gaps in training and awareness can lead to communication breakdowns and make it more difficult for staff to provide effective support.

3.2. Solutions categorized by types

In this section, we have categorized the possible solutions to the above barriers and issues into the following four categories.

1. Information accessibility,
2. Environment,
3. Vehicle, and
4. Staff training.

3.2.1. Information accessibility

Two studies described the use of visual information systems as a way to improve access to travel information for passengers with hearing impairment (Bettger and Pearson, 1989; Hunter-Zaworski and Hron, 1999). It was suggested that large-font, high-contrast displays and real-time updates on departures, delays, and platform changes could improve the situation. These relatively small changes and improvements could reduce reliance on spoken announcements and provide an additional channel for accessing information during travel.

3.2.2. Environment

Two studies described environmental modifications aimed at improving navigation and access to information in transport settings (Fürst and Vogelauer, 2012; Sze and Christensen, 2017). For example, the authors suggested tactile markers and pathways with contrasting colours to support orientation in transit hubs and within vehicles. Furthermore, visual emergency alerts and vibration-based notifications was suggested as alternatives to sound-based warnings. In addition, combining spoken announcements with visual displays and adjusting volume levels in noisy environments was described as a way to improve access to time-sensitive information. In addition, research by Lee. D (2025) in Korea shows that walking satisfaction and pedestrian safety affect vulnerable groups differently. This means that safe and accessible pedestrian spaces are as important as improvements to vehicles (Lee D., 2025).

3.2.3. Vehicle

One study discussed changes to vehicle design intended to improve access to information during travel (Starzyńska et al., 2015). Examples of vehicle design were the use of interior visual displays providing real-time information about current and upcoming stops. Starzyńska et al. (2015) also described seating layouts that allow for clearer pathways and designated spaces for passengers with disabilities, which can make it easier to move within the vehicle and maintain orientation.

3.2.4. Interactivity and personnel training

Four studies addressed the importance of better communication between passengers and transit staff, with a focus on practical ways to support interaction (Bettger and Pearson, 1989; Fürst and Vogelauer, 2012; Hunter-Zaworski and Hron, 1999; Alçura, 2025). The studies suggested simple tools for written communication, such as paper and pens, as well as digital devices that enable real-time message exchange. The studies also emphasized staff and drivers' training, including basic sign language, awareness of hearing-related needs, and familiarity with assistive technologies to recognize the needs of travellers with hearing impairment. Together, these measures were described as ways to reduce communication breakdowns and support more effective assistance in everyday travel situations.

3.3. Visualisation of trip stages

To illustrate how accessibility challenges arise across different parts of a public transport journey, we developed a figure (see Fig. 2) that brings together the barriers and corresponding measures identified in the review. The figure organises these across key stages of travel: pre-trip planning, at the station or stop, boarding, during the journey, and alighting. Structuring the material in this way makes it possible to see how accessibility needs change over the course of a journey, and where particular barriers tend to occur. It also helps identify stages where specific measures may have the greatest impact, depending on how information is accessed and how passengers interact with the transport environment.

3.4. Safety in public transportation

Ensuring the safety of passengers with hearing impairment in public transportation remains a key issue, particularly in environments where communication is largely based on auditory signals. Limited access to spoken information, such as emergency

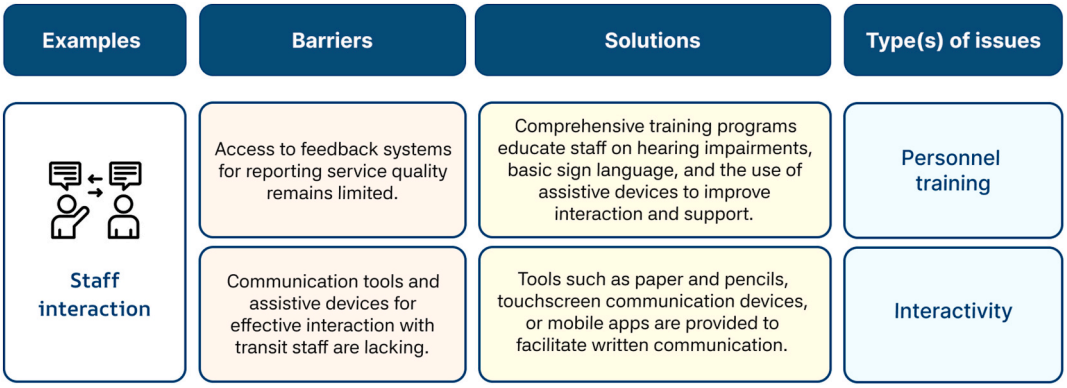


Fig. 3. Barriers and solutions for staff interactions.

announcements or service updates, can increase the risk that important information is missed. This section reviews safety-related measures reported in studies from the United States, United Kingdom, Hong Kong, Austria, Poland, and Sweden, and compares how different systems address these challenges.

In the United States, the Americans with Disabilities Act (ADA) provides a regulatory framework for supporting passengers with hearing impairment (United States Congress, 1990). Reported measures include visual emergency alarms, electronic reader boards, assistive listening devices, captioning, and staff training related to communication and disability awareness (Bettger and Pearson, 1989; United States Congress, 1990). Together, these measures are intended to support access to safety-critical information during travel. In the United Kingdom and Hong Kong, studies describe similar approaches, including audio signals for vehicle arrivals, clear visual signage, induction loop systems (that magnetically improves sound signals sent to cochlear implants or hearing aids), and integrated audio-visual information systems (Sze and Christensen, 2017). These multimodal systems allow information to be accessed through both visual and auditory channels.

In European countries, such as Austria and Poland, the reported measures are more closely tied to general accessibility rather than safety-specific interventions. In Austria, this includes barrier-free design features such as clear announcements, visual information systems, and staff training to improve communication (Fürst and Vogelauer, 2012). In Poland, comparable measures have been described, including auditory information systems, accessible signage, and driver training programmes focused on interaction with passengers with hearing impairment (Starzyńska et al., 2015). In Sweden, recent developments have focused on the improvement of accessibility to real-time information in public transport. For example, in larger cities such as Stockholm, electronic displays provide updates on schedules, delays, and platform changes, reducing reliance on spoken announcements (Stjernborg, 2019). For emergency communication, Sweden offers multiple accessible platforms, including Krisinformation. se and the SOS Alarm 112 application, both providing real-time visual alerts (Stjernborg, 2019). Sweden further offers access to emergency services through 24-h video relay services. Importantly, this service allows individuals with hearing impairment to communicate in Swedish Sign Language via interpreters connected to emergency dispatch centres (Warnicke, 2019).

Despite these developments, the reviewed studies point to remaining challenges, including variation in the availability of visual information across transport modes and differences in staff preparedness related to disability awareness (Stjernborg, 2019; Warnicke, 2019). These findings suggest that further work is needed to ensure that accessibility measures are implemented consistently and support safe travel across the full public transport system.

However, accessibility is not the only component that contributes to safety. Tyler (2017, p. 58) explicitly includes security as a sub-branch of safety defining it as "the state engendered by society so that people can move freely without risk caused deliberately by the actions of others". Recent research has also found that security was among the key concerns of disability groups along with crowdedness, noise and ventilation (Alçura, 2025). This also impacted their satisfaction level with service quality score gaps at −1.42 (Feeling safe at stops (regarding crime)) and −1.31 (In-vehicle safety (crime)). Feeling unsafe or uncomfortable may negatively affect the willingness to travel. Therefore, policies should address not only improving physical convenience but also psychological satisfaction (Lee, 2025). One example is the Travel training launched by Calgary Transit in 1998, a free program that covered trip planning, using Calgary Transit resources, and safety and security, while providing elderly and disabled persons with direct experience on the transit system. (Iannuzziello, 2001). Implementing interventions that improve the security and comfort dimensions are thus essential for increasing use of public transport and social participation among people with disabilities.

3.5. Sustainable services in public transportation

Sustainable services in public transportation must simultaneously address safety, accessibility, and environmental concerns to meet the needs of current and future generations (Tyler, 2017). Social sustainability emphasizes designing inclusive systems that cater to diverse user groups, particularly individuals with reduced mobility, ensuring that transport services are accessible to all segments of

society (Tyler, 2017). Economic sustainability, while less explicitly highlighted, is reflected in approaches that aim to optimize resource use, such as the integration of open data to improve routing efficiency for people with limited mobility (Weyrer et al., 2014). Environmentally, sustainability involves minimizing the ecological impact of transport by promoting efficient routing and reducing unnecessary travel burdens, which is partly supported through multimodal trip planning tools that encourage the use of public and non-motorized transport options (Weyrer et al., 2014). Despite these advancements, challenges remain in ensuring that transport systems are not only technically functional but also socially inclusive and adaptable over time (Tyler, 2017, Pereira et al., 2017). Achieving a truly sustainable transport system thus requires continuous integration of social, economic, and environmental strategies, informed by technological innovation and user-centered design.

3.6. Suggested strategies for improvement

The reviewed studies describe several strategies to improve accessibility in public transportation.

3.6.1. Improved staff training

Three studies mentioned that training transit personnel to better understand the needs of passengers with hearing impairments was considered important (Bettger and Pearson, 1989; Hunter-Zaworski and Hron, 1999; Alçura, 2025). For example, the researchers suggested sensitivity training focused on recognizing hearing impairments, allowing personnel to use basic sign language, and using other communication tools such as written notes and assistive technologies. However, as Tyler (2017) also discussed, the driver might have many other competing objectives that require attention. In addition, it was noted that any training programs implemented also must be periodically updated to reflect technological advances and incorporate user feedback to remain relevant.

In Fig. 3, the barriers and possible solutions in this specific situation are indicated.

3.6.2. Enhanced communication systems

Another area of improvement that two studies suggested was the improvement of the communication systems (Starzyńska et al., 2015; Sze and Christensen, 2017). The authors suggested that the communication systems could make use of synchronized visual and auditory communication methods and mentioned electronic reader boards, captioned displays, and visual emergency alarms. All of these could provide real-time updates on schedules, delays, and emergencies. Moreover, other suggestions included the use of vibration-based notifications for vehicle arrivals and emergency alerts can also enhance accessibility for passengers with hearing impairment.

3.6.3. Universal design and standardization

Importantly, two studies also suggested the adoption of universal design principles and how important such principles is to create inclusive and accessible public transit systems (Fürst and Vogelauer, 2012; Starzyńska et al., 2015). For example, the use of standardized signage with large fonts, high contrast, and tactile pathways was suggested as improvements of navigation for passengers with sensory impairments (e.g. hearing impairment). The implementation of universal solutions to a system useable for everyone.

3.6.4. Integration of assistive technologies

As further developed in the discussion (i.e., Section 4.1), advanced assistive technologies such as induction loop systems, FM/infrared devices, and mobile applications can provide accessible solutions for hearing-impaired passengers. Assistive technologies can allow passengers to receive the information they need directly through hearing aids or real-time visual notifications. Moreover, mobile apps that offer vibration alerts, route-planning tools, and updates on delays or schedule changes were suggested as particularly beneficial for improving accessibility (Bettger and Pearson, 1989; Hunter-Zaworski and Hron, 1999).

4. Discussion

We have considered accessibility, in the context of mobility, to refer to the ability to reach valued destinations for various purposes. Making public transportation useable and accessible for a wider range of users is closely linked to both sustainability goals and efforts to reduce transport-related social exclusion. Increasing accessibility for groups with specific needs can also benefit a broader range of users by making services easier to use and more robust to variation in physical and perceptual abilities.

When examining previous studies about disabilities in general, our aim has been to pick out hearing disabilities specifically. In the surveyed resources we found very little research on this topic specifically. In Park et al. (2023), only two out of 115 articles, and in Shen et al. (2023), one source out of 66 mentioned hearing impairments specifically. This indicates that hearing has received comparatively little attention within the broader scope of accessibility in public transport research, despite its relevance to how information is accessed and used during travel.

We were especially interested in previous knowledge on how to design mobility services for people with hearing impairments. To this end, the literature survey has identified several barriers and associated solutions related to information accessibility, environment, vehicle and interactivity/personnel training. However, to identify current and practically relevant solutions, we extended the review to include recent technological developments beyond the core set of studies, focusing on tools that may support accessibility for people with hearing impairment.

4.1. Available technologies for supporting hearing-impaired passengers

Hearing aids are widely used as a rehabilitation technology in many countries and play a central role in how passengers with hearing impairment access auditory information in public transport. Technologies such as induction loop systems, which generate a magnetic field that can be picked up by the telecoil in hearing aids, allow users to receive audio signals directly. This can improve signal clarity and reduce the impact of background noise in environments such as buses and transit hubs (Bettger and Pearson, 1989). FM and infrared systems have also been used to transmit announcements directly to hearing aids, providing an alternative to relying on ambient sound in the environment (Bettger and Pearson, 1989). At the same time, access to information does not depend solely on hearing aids. Visual communication tools, including captioning and electronic reader boards, provide an additional channel through which travel information can be accessed (Iannuzziello, 2001). Together with staff training in communication and disability awareness, these measures can reduce communication breakdowns and improve access to information during travel (Bettger and Pearson, 1989).

4.1.1. Inclusive services utilising digital technologies

Based on the reviewed literature, a service design perspective is largely absent, particularly in relation to how different components of the transport system are experienced across a full journey. This is also in line with Nielsen (2024), who showed that regulatory approaches to universal design in transport tend to emphasise physical design and visible disabilities, while groups in need of service-oriented solutions receive less attention. To make use of recent technological developments in an inclusive way, universal design principles can be applied not only to individual components, but to the design of transport services as a whole, potentially enabling more flexible ways of travelling. This may include the use of automated or on-demand transport services, which could complement existing public transport and offer alternative travel options for users with specific accessibility needs. Digital and online systems may also support better integration between transport modes, for example by enabling more flexible access to information and coordination across different stages of a journey.

Similarly to universal design approaches, also the transport justice research field (Lucas, 2012) argues that accessibility is not only about finding solutions for minorities, but that it rather can be seen as a way to design and evaluate whole transport services and policies. This is a shift away from individual impairments toward a focus on how to distribute access to transport systems and opportunities for all. Service design might be especially appropriate in identifying systemic barriers and solutions across service journeys and provider ecosystems.

4.2. Advancements in hearing technologies for public transportation

Traditional technologies such as induction loops and FM or infrared systems have long been used to support access to auditory information for passengers with hearing impairment. However, their effectiveness depends on consistent implementation and on environments where auditory signals remain useable. This can limit their usefulness in complex or noisy transport settings. More recent developments, including AI-based tools, have been proposed as ways to extend how information can be accessed in these contexts (see Patel et al., 2025 for a review). One line of development concerns speech-to-text systems that convert spoken announcements into written output in real time. Such systems can make platform information, delays, and service changes more accessible without relying on auditory perception. Related approaches include AI-supported processing in hearing devices, in which algorithms are used to improve sound recognition or to separate speech from background noise (Fabry and Bhowmik, 2021). Another option is the use of captioning systems and electronic displays, which provide a complementary visual channel (e.g., May et al., 2025). Tactile notification systems represent another approach. Wearable devices and mobile applications can provide vibration-based alerts for delays, service disruptions, or emergency messages. These systems offer an alternative to both auditory and visual signals and may be particularly useful in situations where attention is divided or visual information is limited (Kshirsagar et al., 2025). There are also early examples of integrated solutions at the system level. The Dutch railway company NS, for example, has, together with researchers, tested AI-based sign language translation that delivers real-time interpretations of announcements, such as platform changes, to users' smartphones (van Gemert et al., 2022). The same approach has been proposed for onboard situations, including delays where spoken announcements may not be accessible. Rather than relying on fixed infrastructure, these systems shift information delivery to personal devices.

5. Conclusions

Within the reviewed areas, the studies describe a range of practical measures, including high-contrast visual signage, real-time information displays, tactile guidance, visual emergency alerts, and more targeted staff training. Taken together, the findings point to the need for transport systems that combine universal design with digital and assistive technologies, while also improving usability and service quality for vulnerable user groups. They also suggest that accessibility can be considered within a broader development of service design and innovation in public transport, including emerging solutions such as automated vehicles and digitally supported travel.

Addressing the identified barriers is likely to improve access to transport, support independent travel, and increase participation in everyday activities for passengers with hearing impairment. At the same time, several challenges remain. Implementation costs, lack of standardization, and variation across transport systems may limit large-scale adoption. In addition, many solutions continue to assume a uniform user group, despite variation in hearing status, communication preferences, and technology use. This points to the need for

approaches that account for different ways of accessing information, and for transport systems that are adaptable to diverse user needs rather than optimised for a single mode of interaction.

The review also indicates that relatively few studies have examined the transport and mobility needs of passengers with hearing impairment. This gap is particularly relevant in the context of emerging transport systems, where design decisions made today may shape accessibility for future users, including those who rely on public transport as a primary mode of mobility. Given ongoing developments in transport systems, including the introduction of automated and connected services, future research should more explicitly examine how these technologies can support this group. Without such work, there is a risk that new systems reproduce existing barriers rather than reducing them. Furthermore, while the reviewed studies address accessibility, safety, and inclusion for passengers with disabilities, a significant limitation across studies is the lack of attention to intersectionality and gender-based concerns. Although more recent research has identified security (Tyler, 2017), psychological satisfaction (Lee, 2025) and crime-related safety (Alçura, 2025) as important dimensions for travel with public transport, little attention is provided to harassment, sexual- or gender-based violence experienced by women and girls with disabilities. This oversight is worth noting as communication barriers may make it difficult to report such instances or seek assistance from staff. Future research should adopt an intersectional lens that considers how disability, gender, security and accessibility shape public transport experiences, willingness to travel and perceptions of safety. Co-designing with women and girls with disabilities could help ensure that proposed solutions and interventions improve not only safety, accessibility and sustainability but also feelings of security, comfort, inclusion and dignity.

Ensuring safe, just and sustainable transport services for people with disabilities requires continued investment in staff training, communication systems, universal and standardized design, the integration of assistive technologies, and equitable policies that address security, inclusion and social participation. These will not only help people with disabilities but also the population overall. Although this may involve substantial costs, it represents a necessary step toward more inclusive and just services. In the longer term, such developments are likely to contribute to improved social participation and quality of life for passengers with hearing impairment.

CRediT authorship contribution statement

Erik Marsja: Writing – review & editing, Validation. **Vanessa Esperance Rodrigues:** Writing – review & editing. **Arezou Mortazavi:** Writing – review & editing, Investigation, Data curation. **Johan Blomkvist:** Writing – original draft, Project administration, Methodology, Investigation, Funding acquisition.

Declaration of competing interest

The authors declare the following financial interests/personal relationships which may be considered as potential competing interests. Johan Blomkvist reports that Sweden's Innovation Agency (Vinnova) funded the project. The other authors report no known competing financial interests or personal relationships that have influenced the work reported in this paper.

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Data availability

Data will be made available on request.

References

Alçura, G.A., 2025. On the road to inclusion: a multifaceted examination of transportation challenges faced by individuals with disabilities. Sustainability (Switzerland) 17 (1). https://doi.org/10.3390/su17010081.

Bascom, G.W., Christensen, K.M., 2017. The impacts of limited transportation access on persons with disabilities' social participation. J. Transport Health 7, 227–234. https://doi.org/10.1016/j.jth.2017.10.002.

Bettger, G.R., Pearson, T.J., 1989. Accommodating deaf and hard-of-hearing persons on public transportation systems in Massachusetts. Transp. Res. Rec. 1209, 16–18.

Bezyak, J.L., Sabella, S.A., Gattis, R.H., 2017. Public transportation: an investigation of barriers for people with disabilities. J. Disabil. Pol. Stud. 28 (1), 52–60. https://doi.org/10.1177/1044207317702070.

Fabry, D.A., Bhowmik, A.K., 2021. Improving speech understanding and monitoring health with hearing aids using artificial intelligence and embedded sensors. Semin. Hear. 42 (3), 295–308. https://doi.org/10.1055/s-0041-1735136.

Figueiredo, E., Eusébio, C., Kastenholz, E., 2012. How *diverse* are tourists with disabilities? A pilot study on accessible leisure tourism experiences in Portugal. Int. J. Tourism Res. 14 (6), 531–550. https://doi.org/10.1002/jtr.1913.

Fürst, E.W.M., Vogelauer, C., 2012. Mobility of the sight and hearing impaired: barriers and solutions identified. Qual. Mark. Res. Int. J. 15 (4), 369–384. https://doi.org/10.1108/13522751211257060.

Guillén, D.R., Soler, J.A.C., Lucas, J.C.S., 2024. Transport accessibility for disabled people in the European Union. Journal of Infrastructure, Policy and Development 8 (4), 4. https://doi.org/10.24294/jipd.v8i4.3445.

Hunter-Zaworski, K., Hron, M.L., 1999. Bus accessibility for people with sensory disabilities. Transp. Res. Rec.: J. Transport. Res. Board 1671 (1), 40–47. https://doi.org/10.3141/1671-07.

Iannuzziello, A.S., 2001. *Communicating with Persons with Disabilities in a Multimodal Transit Environment* (No. 37; Transit Cooperative Research Program (TCRP) Synthesis). Transportation Research Board.
Kshirsagar, U., Oh, T., Tomaszewski, B., Behm, G., 2025. Inclusive Emergency Alerts: a Cloud- Based Haptic and Visual Notification System for the Deaf and Hard of Hearing.
Lee, D., 2025. The impact of the walking environment on the use of public transportation by the transportation vulnerable. KSCE J. Civ. Eng. 29 (5), 100094. https://doi.org/10.1016/J.KSCEJ.2024.100094.
Lubin, A., Deka, D., 2012. Role of public transportation as job access mode: lessons from survey of people with disabilities in New Jersey. Transp. Res. Rec. 2277 (1), 90–97. https://doi.org/10.3141/2277-11.
Lucas, K., 2012. Transport and social exclusion: where are we now? Transport Policy, Urban transport initiatives 20, 105–113. https://doi.org/10.1016/j.tranpol.2012.01.013.
May, L., Clemens, M., Dang, K., Ohshiro, K., Sridhar, S., Wee, P., Fuentes, M., Lee, S., Cartwright, M., 2025. "Choices? That's the dream": challenges and opportunities in non-speech information closed-captioning. Front. Comput. Sci. 7, 1575176. https://doi.org/10.3389/fcomp.2025.1575176.
Nielsen, A.F., 2024. How laws of universal design discriminate between different types of disabilities—Lessons learned from Norway. J. Transport Health 37, 101821. https://doi.org/10.1016/j.jth.2024.101821.
Park, J., Chowdhury, S., 2018. Investigating the barriers in a typical journey by public transport users with disabilities. J. Transport Health 10, 361–368. https://doi.org/10.1016/j.jth.2018.05.008.
Park, K., Esfahani, H.N., Novack, V.L., Sheen, J., Hadayeghi, H., Song, Z., Christensen, K., 2023. Impacts of disability on daily travel behaviour: a systematic review. Transp. Rev. 43 (2), 178–203. https://doi.org/10.1080/01441647.2022.2060371.
Patel, P., Pampaniya, S., Ghosh, A., Raj, R., Karuppaih, D., Kandasamy, S., 2025. Enhancing accessibility through machine learning: a review on visual and hearing impairment technologies. IEEE Access 13, 33286–33307. https://doi.org/10.1109/ACCESS.2025.3539081.
Pereira, R.H.M., Schwanen, T., Banister, D., 2017. Distributive justice and equity in transportation. Transp. Rev. 37 (2), 170–191. https://doi.org/10.1080/01441647.2016.1257660.
Rimmer, J.H., Riley, B., Wang, E., Rauworth, A., Jurkowski, J., 2004. Physical activity participation among persons with disabilities. Am. J. Prev. Med. 26 (5), 419–425. https://doi.org/10.1016/j.amepre.2004.02.002.
Scott, J., D'Arcangelo, M., Olness, B., Grimm, M., Bush, T., 2023. An assessment of current barriers to accessibility in public transportation pick up/drop off zones and how solutions may be applied to autonomous vehicles. WCX SAE World Congress Experience. https://doi.org/10.4271/2023-01-0713.
Shen, X., Zheng, S., Wang, R., Li, Q., Xu, Z., Wang, X., Wu, J., 2023. Disabled travel and urban environment: a literature review. Transport. Res. Transport Environ. 115, 103589. https://doi.org/10.1016/j.trd.2022.103589.
Starzyńska, B., Kujawińska, A., Grabowska, M., Diakun, J., Więcek-Janka, E., Schnieder, L., Schlueter, N., Nicklas, J.-P., 2015. Requirements elicitation of passengers with reduced mobility for the design of high quality, accessible and inclusive public transport services. Manag. Prod. Eng. Rev. 6 (3), 70–76. https://doi.org/10.1515/mper-2015-0028.
Stjernborg, V., 2019. Accessibility for all in public transport and the overlooked (social) dimension—A case study of Stockholm. Sustainability 11 (18), 4902. https://doi.org/10.3390/su11184902.
Sze, N.N., Christensen, K.M., 2017. Access to urban transportation system for individuals with disabilities. IATSS Res. 41 (2), 66–73. https://doi.org/10.1016/j.iatssr.2017.05.002.
Trafikanalys, 2018. Resvanor och funktionsnedsättningar—Statistik ur resvaneundersökningen. https://www.trafa.se/kollektivtrafik/resvanor-och-funktionsnedsattningar---statistik-ur-resvaneundersokningen-7993/.
Tyler, N., 2017. Safety accessibility and sustainability: the importance of micro-scale outcomes to an equitable design of transport systems. IATSS Res. 41 (2), 57–65. https://doi.org/10.1016/j.iatssr.2017.06.002.
UN, 2006. Convention on the rights of persons with disabilities (CRPD) | division for inclusive social development (DISD). https://social.desa.un.org/issues/disability/crpd/convention-on-the-rights-of-persons-with-disabilities-crpd.
United States Congress, 1990. Americans with Disabilities Act.
van Gemert, B., Cokart, R., Esselink, L., de Meulder, M., Sijm, N., Roelofsen, F., 2022. First steps towards a signing avatar for railway travel announcements in the Netherlands. Proceedings of the 7th International Workshop on Sign Language Translation and Avatar Technology (SLTAT 7), 109–116. https://aclanthology.org/2022.sltat-1.14.
Warnicke, C., 2019. Equal access to make emergency calls: a case for equal rights for deaf citizens in Norway and Sweden. Soc. Incl. 7 (1), 173–179. https://doi.org/10.17645/si.v7i1.1594.
Weyrer, T.N., Hochmair, H.H., Paulus, G., 2014. Intermodal door-to-door routing for people with physical impairments in a web-based, open-source platform. Transp. Res. Rec.: J. Transport. Res. Board 2469 (1), 108–119. https://doi.org/10.3141/2469-12.
WHO, 2023. Disability. https://www.who.int/news-room/fact-sheets/detail/disability-and-health.